

Erdős Problem #41

EP-41

L1 - TRACTABLE

OPEN

Combinatorics

ORIGINAL PROBLEM STATEMENT

Let $A \subset \mathbb{N}$ be an infinite set such that the triple sums $a + b + c$ are all distinct for $a, b, c \in A$ (aside from the trivial coincidences). Is it true that $\liminf \frac{|A \cap \{1, \dots, N\}|}{N^{1/3}} = 0$?

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Archive: <https://www.unsolvedmath.com/problems?difficulty=1>
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>